

zkBugs Evaluation Results

Table 1: Tool Performance Summary

Tool	TP	FN	Timeout	Failure	Total	Median Time (s)
circumspect	0	70	0	0	70	0.03
circum_civer	12	11	22	25	70	0.88
picus	10	35	19	6	70	2.62
ecneproject	25	38	6	1	70	22.21
zkfuzz	25	16	25	4	70	0.97
conscs	11	47	11	1	70	2.46
TOTAL	83	217	83	37	420	1.97

Table 2: Execution Time Statistics

Tool	Total	Min (s)	Max (s)	Median (s)	Mean (s)	NR-Timeout
circumspect	69	0.01	0.33	0.03	0.07	0
circum_civer	23	0.04	502.33	0.88	86.00	22
picus	51	1.08	494.72	2.62	77.59	19
ecneproject	63	12.93	148.34	22.21	30.10	6
zkfuzz	41	0.01	531.69	0.97	40.41	25
conscs	58	0.25	491.43	2.46	56.48	11

Table 3: Bug-Tool Matrix (by Bug ID)

Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz
1 FN	FN	FN	FN	FN	FN
2 Timeout	FN*	Timeout	FN	Timeout	Timeout
3 FN	FN	Timeout	Timeout	TP*	TP*
4 FN	FN	FN	FN	FN	TP*
5 FN	FN	FN	Failure	FN	TP*
6 FN	FN	Failure	FN	FN	Failure
7 TP*	FN	Failure	Timeout	TP*	Failure
8 FN	FN*	Failure	FN	FN	Failure
9 FN	FN	TP	TP*	TP*	TP*
10 FN	FN*	Timeout	Timeout	TP*	TP*
11 TP*	FN	TP	TP*	TP*	TP*
12 FN	FN	Timeout	FN	FN	Timeout
13 FN	FN	Timeout	FN	FN	Timeout
14 Timeout	FN	FN	Timeout	FN	TP*
15 FN	FN	FN	Timeout	FN*	Timeout
16 FN	FN	FN	Timeout	FN	TP*
17 FN	FN	Timeout	Timeout	TP*	FN
18 FN	FN	FN	FN*	FN*	Timeout
19 Timeout	FN	FN	FN	FN	Timeout
20	FN	Timeout	Timeout	Timeout	FN

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Bug ID	circumspect	circom_civer	picus	ecneproject	zkfuzz
FN					
21	FN*	Timeout	FN	Timeout	Failure
FN					
22	FN*	Timeout	Timeout	Timeout	FN
FN					
23	FN*	Timeout	Timeout	FN	Timeout
FN					
24	FN*	TP	TP*	TP*	TP*
TP*					
25	FN	Failure	FN	FN	FN
FN					
26	FN*	Failure	FN*	FN*	Timeout
FN					
27	FN*	Failure	FN	FN	FN
FN					
28	FN	Failure	FN	FN	Timeout
FN					
29	FN*	Failure	FN*	FN*	Timeout
FN					
30	FN	TP	TP*	TP*	TP*
TP*					
31	FN*	Failure	FN*	FN*	TP*
FN					
32	FN*	FN	FN*	FN*	FN*
FN					
33	FN	Failure	FN*	Timeout	FN
Timeout					
34	FN	Timeout	Timeout	TP*	Timeout
FN					
35	FN	Timeout	FN	FN	Timeout
FN					
36	FN	FN	FN	FN	Timeout
Timeout					
37	FN	Failure	FN	Timeout	FN
Timeout					
38	FN	Failure	FN	FN	Timeout
FN					
39	FN	Timeout	FN*	FN*	Timeout
FN					

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Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz
40 FN Timeout	FN*	FN	FN	FN	Timeout
41 Timeout	FN*	Timeout	Timeout	TP*	TP*
42 FN	FN	Timeout	Timeout	FN	Timeout
43 FN	FN	TP*	Failure	TP*	TP*
44 FN	FN	TP*	Failure	TP*	TP*
45 FN	FN	TP*	Failure	TP*	TP*
46 TP*	FN	TP	TP*	TP*	TP*
47 TP*	FN	TP	TP*	TP*	TP*
48 TP*	FN	TP	TP*	TP*	TP*
49 Timeout	FN	Timeout	TP*	TP*	TP*
50 FN	FN*	Failure	FN	FN	Timeout
51 FN	FN	Failure	FN	FN	FN
52 FN	FN	Failure	FN	FN	FN
53 TP*	FN*	Timeout	Timeout	TP*	FN
54 FN	FN	Failure	FN	FN	TP*
55 Timeout	FN	Failure	FN	Failure	FN
56 FN	FN	Timeout	FN	FN	Timeout
57 Timeout	FN	Timeout	Timeout	TP*	Timeout
58 FN	FN	Timeout	Timeout	TP*	TP*
59	FN*	TP	TP*	TP*	TP*

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Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz
TP*					
60	FN*	Timeout	Failure	TP*	TP*
TP*					
61	FN	Failure	Timeout	TP*	Timeout
FN					
62	FN*	Failure	FN*	FN*	FN*
FN					
63	FN*	Failure	FN*	FN*	FN*
FN					
64	FN	Failure	FN	FN	FN
FN					
65	FN*	Failure	Failure	FN	Timeout
FN					
66	FN	Failure	Timeout	TP*	Timeout
Failure					
67	FN	Failure	FN	FN	Timeout
FN					
68	FN	Timeout	Timeout	FN	Timeout
Timeout					
69	FN*	Failure	FN	FN	TP*
FN					
70	FN*	TP	TP*	TP*	TP*
TP*					

* = *manually analyzed*

Table 4: Bug ID to Name Mapping

ID	Bug Name
1	daira_hopwood_darkforest_v0_3_missing_bit_length_check
2	hashcloak_data_are_not_fully_verified_during_state_update
3	kobi_gurkan_mimc_hash_assigned_but_not_constrained
4	koukyosyumei_checkdepth_comparator_overflow
5	koukyosyumei_clamp_comparator_overflow
6	koukyosyumei_emitifinrange_lessthan_overflow
7	koukyosyumei_extractstringfrompoint_shiftedfirstbyte_unconstrained
8	koukyosyumei_spend_missing_range_check
9	koukyosyumei_subtract_unconstrained_multiplier
10	koukyosyumei_unconstrained_base64_decoded_len
11	koukyosyumei_under_constrained_date_encoder
12	leastauthority_Circuit_Does_Not_Check_the_ERC_20_Sum_Correctly_
13	leastauthority_previously_correct_ownership_proof_disabled_via_code_changes
14	matterlabs_email_spoofing_via_manipulated_from_header
15	matterlabs_emailauth_fails_with_overlapping_invitation_code_regex
16	matterlabs_fromaddrregex_allows_email_address_spoofing
17	matterlabs_underconstrained_fpmul_circuit
18	openzeppelin_mismatched_base64url_decoding_may_break_completeness
19	openzeppelin_non_determinism_of_some_inputs
20	trailofbits_incorrect_handling_of_point_doubling_can_allow_signature_forgery
21	trailofbits_prover_can_lock_user_funds_by_including_ill-formed_bigints_in_public_key_commitment
22	trailofbits_prover_can_lock_user_funds_by_supplying_non-reduced_Y_values_to_G1BigIntToSignFlag
23	trailofbits_unsafe_use_of_num2bits_in_multiple_circuits
24	veridise_arrayxor_is_under_constrained
25	veridise_babyjubjub_suborder_constraints_not_applied_correctly
26	veridise_blacklist_states_not_representable_in_field
27	veridise_bypassing_internal_transfer_limits_via_swaps
28	veridise_creditnoteproofs_can_be_stolen
29	veridise_data_escrow_encrypted_message_wrong_input
30	veridise_decoder_accepting_bogus_output_signal
31	veridise_fortxreward_abstracted_away_in_reward_calc
32	veridise_incorrect_initialization_in_membership_circuits
33	veridise_kyt_signature_verification_fails_nonzero_hash
34	veridise_missing_range_checks_in_bigmod
35	veridise_missing_range_checks_on_comparison_circuits
36	veridise_no_zero_value_validation
37	veridise_nullifier_verification_can_be_disabled
38	veridise_ownershipproofs_could_identify_issuers
39	veridise_private_information_leakage

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ID	Bug Name
40	veridise_spammers_may_slash_themselves
41	veridise_template_CoreVerifyPubkeyG1_does_not_perform_input_validation_simplified
42	veridise_underconstrained_circuit_allows_invalid_comparison
43	veridise_underconstrained_outputs_in_bitElementMulAny
44	veridise_underconstrained_outputs_in_window4
45	veridise_underconstrained_outputs_in_windowmulfix
46	veridise_underconstrained_points_in_edwards2Montgomery
47	veridise_underconstrained_points_in_montgomery2Edwards
48	veridise_underconstrained_points_in_montgomeryAdd
49	veridise_underconstrained_points_in_montgomeryDouble
50	veridise_unsafe_num2bits_254_blacklist_leaf
51	veridise_zaccount_renewal_kyc_expiry_not_validated
52	veridise_zaccount_renewal_multiple_nullifiers_same_utxo
53	veridise_zero_padding_for_sha256_in_ExpandMessageXMD_is_vulnerable_to_an_overflow
54	veridise_zone_id_inclusion_prover_bypass
55	veridise_zone_limits_bypass_zswapv1
56	yacademy_Knowledge_of_any_member_signature_allow_to_generate_proof_of_membership
57	yacademy_input_signal_s_is_not_constrained_in_eff_ecdsa_circom
58	yacademy_under_constrained_circuits_compromising_the_soundness_of_the_system
59	zksecurity_Unsound_Addition_Gadget
60	zksecurity_Unsound_XOR_gadget
61	zksecurity_an_attacker_can_craft_a_fake_non_inclusion_proof_for_a_given_key_due_to_an_aliasing_bug_in_the_smt_verifier
62	zksecurity_big_integer_zero_check_is_not_sound
63	zksecurity_exclusion_check_of_forbidden_countries_is_unsound_and_incomplete_due_to_incorrect_indexing
64	zksecurity_forbidden_country_check_bypass_via_packed_byte_overflow
65	zksecurity_missing_boolean_constraints_in_the_merkle_tree_path_leads_to_an_attacker_being_able_to_craft_a_fake_merkle_proof_for_an_arbitrary_L
66	zksecurity_missing_byte_range_checks_allows_packed_data_pollution
67	zksecurity_second_pre_image_attacks_on_packbytesandposeidon_may_be_used_to_register_arbitrary_passports_and_dsc_certificates
68	zksecurity_sha256_templates_return_zero_on_arbitrary_inputs
69	zksecurity_the_registration_and_disclosure_circuits_lack_range_checks_for_the_input_indices
70	zksecurity_unsound_left_rotation

Table 5: Execution Times per Bug (by Bug ID, in seconds)

Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz
1 0.28	0.02	0.15	1.85	22.80	89.63
2 —	0.30	—	333.50	—	—
3 9.79	0.01	—	—	20.73	0.04
4 0.31	0.01	0.14	1.19	13.40	0.03
5 0.74	0.01	0.13	238.67	23.09	0.02
6 0.28	0.01	—	1.48	17.50	—
7 3.10	0.01	—	—	18.17	—
8 20.10	0.02	—	2.61	27.04	—
9 0.25	0.01	0.07	2.31	25.55	0.02
10 1.97	0.03	—	—	26.94	0.13
11 0.27	0.01	0.29	2.62	23.19	2.63
12 59.04	0.18	—	10.77	48.44	—
13 7.13	0.16	—	4.36	32.31	—
14 —	0.02	66.86	—	20.95	1.38
15 254.47	0.02	249.13	—	18.93	—
16 491.43	0.02	49.64	—	16.06	0.97
17 29.00	0.02	507.20	—	23.56	15.01
18 27.25	0.04	0.88	446.72	23.64	—
19 —	0.02	64.33	21.59	76.66	—
20	0.06	—	—	—	42.87

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Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz
208.95					
21	0.18	—	2.07	—	2.45
0.42					
22	0.23	—	—	—	9.71
82.74					
23	0.01	500.26	—	19.82	—
0.39					
24	0.06	0.13	1.32	15.82	0.12
0.26					
25	0.10	—	3.48	29.35	6.79
0.27					
26	0.13	—	2.94	22.41	—
2.97					
27	0.09	—	3.53	28.22	461.40
0.29					
28	0.02	—	3.85	22.89	—
15.29					
29	0.26	—	17.80	91.09	—
132.52					
30	0.01	0.04	1.35	18.08	0.09
0.40					
31	0.10	—	1.87	21.03	0.10
0.32					
32	0.04	0.06	1.98	27.61	16.65
1.66					
33	0.32	—	444.92	—	39.30
—					
34	0.03	—	—	27.53	—
166.43					
35	0.01	504.09	1.16	12.93	—
4.63					
36	0.19	13.99	4.85	34.13	—
—					
37	0.29	—	439.58	—	40.47
—					
38	0.07	—	1.11	13.70	—
0.25					
39	0.11	—	7.91	44.08	—
40.10					

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Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz
40 1.03	0.10	4.02	1.54	15.08	—
41 39.53	0.24	501.91	—	34.89	0.86
42 33.96	0.03	553.48	—	18.06	—
43 107.65	0.02	502.19	81.75	16.91	0.05
44 20.84	0.04	502.33	338.44	20.02	3.22
45 20.90	0.02	502.30	338.24	20.33	1.69
46 0.31	0.01	0.09	1.59	18.91	0.41
47 0.35	0.01	0.04	1.34	16.19	0.08
48 0.86	0.01	2.06	1.43	16.31	0.01
49 —	0.01	500.10	1.48	16.74	0.01
50 2.96	0.15	—	3.45	25.29	—
51 481.27	0.26	—	55.23	141.61	86.32
52 482.35	0.24	—	56.28	142.66	99.27
53 0.35	0.06	500.16	—	22.96	531.69
54 0.30	0.04	—	1.53	18.87	0.16
55 —	0.33	—	454.02	328.75	41.22
56 6.65	0.05	—	3.88	33.87	—
57 —	0.04	—	—	24.90	—
58 25.08	0.03	—	—	25.84	0.22
59	0.01	0.07	1.69	22.30	0.02

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Bug ID	circomspect	circom_civer	picus	ecneproject	zkfuzz
0.27 60	0.01	500.26	494.72	22.21	0.03
0.33 61	0.01	—	—	19.49	—
489.04 62	0.01	—	1.08	14.07	17.04
0.27 63	0.01	—	1.08	15.92	71.24
0.33 64	0.03	—	1.47	17.31	74.86
0.33 65	0.01	—	104.05	18.91	—
33.37 66	0.03	—	—	27.95	—
66.24 67	0.02	—	1.60	13.21	—
3.12 68	0.04	—	—	148.34	—
— 69	0.01	—	1.47	17.48	0.02
0.29 70	—	19.04	2.53	22.17	0.84
0.26					